

# Chapter 8 — Creating effective documentation

With documentation components named, the next question is how to write them well

# Learning objectives

- Recognize a compact multi-section documentation template

# A four-move documentation workflow

- Effective documentation is iterative across the dataset lifecycle
- Four moves organize the work

## Example 8.13 — Documentation Depth by Intended Use

- Example 8.13 — hands-on module
- Example 8.13 contrasts depth for machine learning training versus exploratory analysis
- Training releases typically need richer preprocessing, label, and split notes
- Purpose decisions drive how much detail each component receives
- Explore the chapter example module
- View files: ``modules/chapter8/example13/``

# Markdown and everyday formats

- For smaller projects, Markdown or plain text is easy to create, share
- Spreadsheets and templates help standardize metadata tables across many datasets
- Larger programs may add automated metadata generation from tools that already track

## Example 8.15 — Markdown Documentation Outline

- Example 8.15 — hands-on module
- Example 8.15 sketches a Markdown layout with overview, table of contents, data dictionary
- Explore the chapter example module
- View files: ``modules/chapter8/example15/``

# Structured formats and FAIR principles

- Structured metadata such as JSON-LD
- Reusable with sufficient methodology, limitations, and licensing detail

## Example 8.19 — Retail Transaction Documentation Template

- Example 8.19 — hands-on module
- Example 8.19 assembles a compact template
- Explore the chapter example module
- View files: ``modules/chapter8/example19/``



# Common documentation challenges

- Weak organizational buy-in
- Treating documentation as a release gate

# Takeaways

- Documentation depth follows intended use
- Maintainable formats range from Markdown to structured Schema.org-style metadata
- FAIR criteria turn completeness into discoverability and reuse
- Templates like the retail example show how overview, dictionary, provenance

# Next

- Complete the quiz for this part
- Documentation freezes meaning at a point in time